

Types of ML

- ① Supervised ML
- ② Unsupervised ML
- ③ Semi Supervised ML
- ④ Reinforcement

① Supervised ML └→ CLASSIFICATION
└→ REGRESSION

Dataset → Specific o/p → Model → o/p

CLASSIFICATION

Independent features

Play hours

Study hours

8

2

7

3

3

6

Pass/Fail

⇒ Categorical feature

Fail

Fail

Pass

↑ o/p feature → dependent feature

Regression : o/p ⇒ continuous

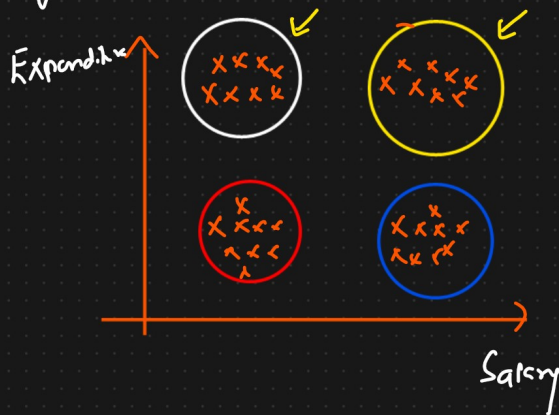
Size of House

No. of Room

Price ^{↑ o/p} ⇒ continuous

② Unsupervised ML : Dataset $\rightarrow$ Clusters or Similar groups

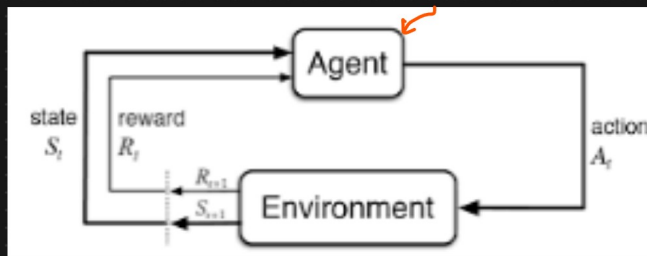
Eg: Customer Segmentation No o/p



Product $\rightarrow$ clusters
 $\downarrow$
Target the audience
 $\Downarrow$
20% Discount

③ Semi Supervised : Supervised + Unsupervised

④ Reinforcement Learning :



Reinforcement learning is an area of machine learning concerned with how intelligent agents ought to take actions in an environment in order to maximize the notion of cumulative reward. Reinforcement learning is one of three basic machine learning paradigms, alongside supervised learning and unsupervised learning.

